Section 8.1 Using symbols

Symbols such as ▲, ● or ◆ are often used to stand for numbers.

Sometimes you can work out the value of the symbol.

Look at this example:

▼ + 8 = 12

We know that 4 + 8 = 12, so this means that ▼ = 4.

You can also use the value of a symbol in a calculation.

If ● = 7, then 6 × ● = 42.

Writing in symbols

'3 more than 5' can be written as 5 + 3.

'6 less than 14' can be written as 14 − 6.

'10 increased by 4' can be written as 10 + 4.

In the same way,

'5 more than ■' can be written as ■ + 5.

'7 more than n' can be written as $n + 7$.

'8 less than x' can be written as $x - 8$.

Exercise 8.1

1. Each symbol stands for a number. Write down its value.

(i) ▲ + 3 = 5 (ii) ■ − 5 = 4 (iii) □ × 2 = 12 (iv) 3 + ◆ = 15
(v) 9 − ● = 2 (vi) 5 × ■ = 55 (vii) ▼ ÷ 3 = 4 (viii) ◆ ÷ 10 = 3

2. Find the value of each of these symbols:

(i) 11 − ◀ = 3 (ii) 14 + ♣ = 21 (iii) ◗ + ◗ = 20
(iv) ✸ + ✸ + ✸ = 30 (v) □ − 6 = 14 (vi) ◆ × ◆ = 16
(vii) ▲ × 4 = 20 (viii) ✸ ÷ 6 = 5

3. If □ + ▲ = 12, find three pairs of values for □ and ▲.

4. If ● − ◆ = 9, find three pairs of values for ● and ◆.

5. If ● = 5 and ■ = 3, find the value of each of these expressions:

(i) ● + 6 (ii) ■ × 5 (iii) ● − 2 (iv) 7 + ■
(v) 3 × ● (vi) 6 ÷ ■ (vii) 30 ÷ ● (viii) 2 × ■ + ●

6. If ▲ = 18, write down the value of each of these:

(i) 6 more than ▲ (ii) 3 less than ▲ (iii) twice ▲
(iv) five times ▲ (v) ▲ less 8 (vi) twice ▲ + ▲

7. ■ stands for different numbers. Find its value in each of the following:

(i) ■ + 6 = 10 (ii) ■ − 3 = 5 (iii) 7 + ■ = 14
(iv) 2 × ■ = 22 (v) ■ ÷ 4 = 9 (vi) 3 × ■ = 36

8. ▲ stands for a number. Write down each of these using symbols and numbers only:

(i) 7 more than ▲ (ii) 4 less than ▲ (iii) twice ▲
(iv) ▲ divided by 2 (v) $\frac{1}{3}$ of ▲ (vi) ▲ less 6

9. The first line in the table below has been completed.
Write down what should go in each shaded space in the remaining lines.

	Start number/symbol	Change	Result
	3	Increase by 7	3 + 7 = 10
(i)	9	Increase by 12	
(ii)	12		12 + 6 = 18
(iii)		Increase by 8	21 + 8 = 29
(iv)	□		□ + 5
(v)	▼	Increase by 8	
(vi)	●	Increase by 17	
(vii)		Decrease by 3	16 − 3 = 13
(viii)		Decrease by 12	■ − 12
(ix)	▲	Decrease by 36	
(x)	◗		◗ − 9